Date:- 27-10-26

231-

eg: input : 1 output: true

$2^0 \rightarrow 1 \rightarrow 1$ can be written in form of 2 power so, 1 is true soln of this.

eg: input : 16 output: 16

$2^4 = 16 \rightarrow$ 16 can be written in form of 2 powers so, 16 is true soln of this.

Approach and logic

A power of two has exactly one set bit(1) in its binary representation

eg. $8 = 1000_2$

$7 = 0111_2$

Code

```
class Solution {
    public boolean isPowerOfTwo(int n) {
        return n > 0 && (n & (n-1)) == 0;
    }
}
```

- n>0 ensuring that number is positive.
- n & (n-1) remove the rightmost set bit
- A power of two has exactly one set bit, so the result must be 0.

Time Complexity : O(1)
Space Complexity : O(1)

Description | Accepted | Editorial | Solutions | Submissions

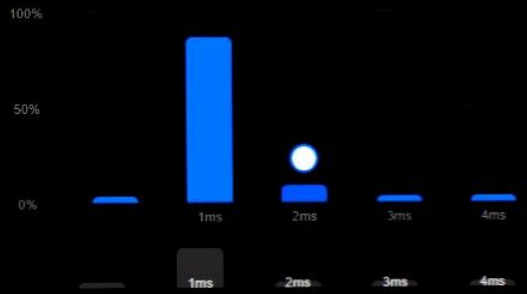
← All Submissions

Runtime

2 ms | Beats **9.98%**

Memory

42.08 MB | Beats 92.83%



Code Java

Code

Java | Auto

```java
class Solution {
    public boolean isPowerOfTwo(int n) {
        return n > 0 && (n & (n-1)) == 0;

    }
}
```

Saved

Ln 3, Col 42

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2 | Case 3

Input

n =

1

Output

true